Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad Cantonment, Natore–6431

Department of Computer Science and Engineering

Course Code: CSE 4105 | Course Title: Data Science and Applications

Total Marks: 15 | Submission Date: 03/11/2026

Assignment: "Impact of Transformations in Modeling and Forecasting with ARIMA"

Submission Instructions

1. **Format (Printed vs. Handwritten):**
   - **Front Cover Page:** Must be printed with student details.
   - **Body Pages:** All solutions, tables, equations, plots, and code excerpts must be handwritten by the student.
2. **Required Tool:** All analytical work, statistical tests, model fitting, and forecasting must be performed in the **R Programming Language**.
3. **Late Submission Penalty:** Submissions after the deadline (November 03, 2026) will incur a **5 marks deduction**.
4. **Dataset Selection Rule (R Modulo Operation):** Select the R dataset based on your Student ID: `ID_last_two_digits %% 6`
   - **Remainder 0** → Use dataset `nottem`
   - **Remainder 1** → Use dataset `USAccDeaths`
   - **Remainder 2** → Use dataset `austres`
   - **Remainder 3** → Use dataset `UKgas`
   - **Remainder 4** → Use dataset `AirPassengers`
   - **Remainder 5** → Use dataset `UKDriverDeaths`

Assignment Tasks & Questions

1. **Exploratory Data Analysis & Data Splitting (2 Marks)**
   Plot the series (use time-plot, ACF plot, and seasonal-subseries plot) in R and describe its main features. Split the data into training (70%) and test (30%) data sets. Use the training data for answering questions 2 and 3, and use the test data for answering question 6.
2. **Stationarity & Original Scale ARIMA Modeling - M1 (2.5 Marks)**
   Take the suitable number of non-seasonal and/or seasonal differences (if required) to get a stationary series.
   Hints: Stationarity can be checked by Augmented Dickey-Fuller Test (`adf.test`) and KPSS Test (`kpss.test`) in the `tseries` package. In KPSS, accepting null means stationary (small p-value suggests differencing is required). In ADF, a small p-value suggests stationary.
   Choose a model using ACF & PACF plots, designated as model **M1**. Fit the M1 model, write its mathematical expression, and perform residual diagnostic checking. Repeat until residual assumptions are fulfilled.

3. **Classical Transformation & ARIMA Modeling - M2 (2.5 Marks)**
   Apply an appropriate transformation (Square root, Cube root, or Logarithm). Take suitable differences to achieve stationarity, select model **M2** using ACF & PACF plots, write its mathematical expression, and perform residual diagnostic checking.

4. **Box-Cox Transformation & ARIMA Modeling - M3 (2.5 Marks)**
   Apply Box-Cox transformation with the optimum λ, take suitable differences to obtain stationarity, select model **M3** using ACF & PACF plots, write its mathematical expression, and perform residual diagnostics.

5. **Auto ARIMA Modeling - M4 & Comparison (1.5 Marks)**
   Using `auto.arima` function, fit model **M4**. Based on your training dataset, compare models **M1**, **M2**, **M3**, and **M4** using AIC, AICc, and BIC in Table 1. Check which model fulfilled more residual assumptions.

Table 1: Model Selection Criteria Comparison

Model	AIC	AICc	BIC
M1			
M2			
M3			
M4			

6. **Out-of-Sample Forecasting (1 Mark)**
   Estimate the forecasted values based on models **M1**, **M2**, **M3**, and **M4**. Back-transforming is required for **M2** and **M3** to get forecasts on the original scale.

7. **Forecast Accuracy Evaluation (1.5 Marks)**
   Evaluate all models on test data using forecasting accuracy measures (MSE, RMSE, MAE, MPE, MAPE). Record results in Table 2 and find the best model.

Table 2: Model Accuracy Measures on Test Dataset

Model	MSE	RMSE	MAE	MPE (%)	MAPE (%)
M1					
M2					
M3					
M4					

8. **Future Prediction (1 Mark)**
   Using the best model, forecast 10 points ahead and plot that.

Evaluation Rubric (15 Marks Total)

— Best of Luck —

Component	Marks	Evaluation Criteria
Task 1: EDA & Splitting	2.0	Correct time-plot, ACF, seasonal subseries plot, feature description, and 70/30 data split.
Task 2: Model M1	2.5	Correct ADF/KPSS test application, differencing, ACF/PACF model choice, mathematical equation, residual diagnostics.
Task 3: Model M2	2.5	Proper choice of classic transformation, differencing, model selection, equation, residual validation.
Task 4: Model M3	2.5	Optimum λ Box-Cox transformation, differencing, model selection, equation, residual diagnostics.
Task 5: Auto ARIMA M4	1.5	Correct `auto.arima` fit, model criteria table (AIC, AICc, BIC) completion, residual evaluation.
Task 6: Forecasting	1.0	Accurate test set forecasts and proper back-transformation for M2/M3.
Tasks 7 & 8: Evaluation & Future Plot	3.0	Computation of error metrics (MSE, RMSE, MAE, MPE, MAPE), best model selection, 10-step future forecast plot.
Late Penalty	**–5.0**	**Applied automatically if submitted after deadline (03/11/2026).**